



Dementia Early Detection

Classical & Quantum Machine Learning on OASIS Brain MRI Dataset

Team Beta: Dalia Mousa , Enas Emad, and Haneen Ghanem

Mentor: Dr. Maha Metawe

OASIS Longitudinal Dataset · PennyLane VQC · Scikit-learn



Team Contribution:

Team Beta

Dalia Mousa: Introduction and Problem Statement

Enas Emad: Classical and Quantum Pipeline

Haneen Ghanem: Business Analysis and Future Work

Mentor: Dr. Maha Metawe



The Problem & Dataset

Why Early Detection?

- Dementia affects 55M+ people worldwide
- Early diagnosis can slow disease progression
- Brain changes occur years before symptoms
- Goal: classify patients into 3 groups using MRI-derived features

OASIS Longitudinal Dataset

150

Subjects (60–96 yrs)

64

Demented

373

MRI Sessions

14

Converted

72

Nondemented

Features: Age · Sex · EDUC · SES · MMSE · CDR · eTIV · nWBV · ASF · MR Delay

Pipeline Overview



1. Preprocess

- Fill nulls (mean)
- Encode labels
- StandardScaler
- Subject-Wise Split
- 70% / 15% / 15%



2. Classical ML

- 8 classifiers
- 5-fold CV
- Acc · F1 · AUC



3. Quantum VQC

- 4 VQC designs
- Angle encoding
- PennyLane Adam
- 100 epochs max
- Early Stopping
- Patience = 5 epochs



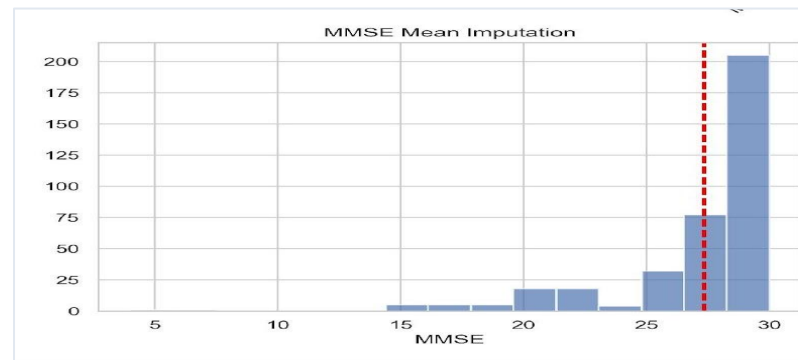
4. Compare

- Classical vs QML
- Metrics table
- Business insights

Preprocessing Insights

Main 10 Features

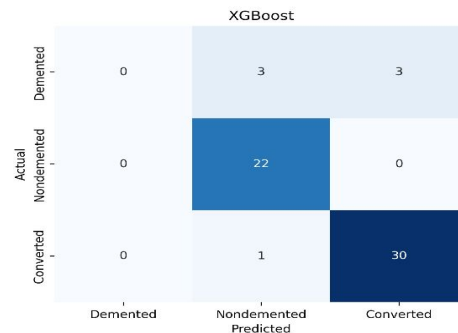
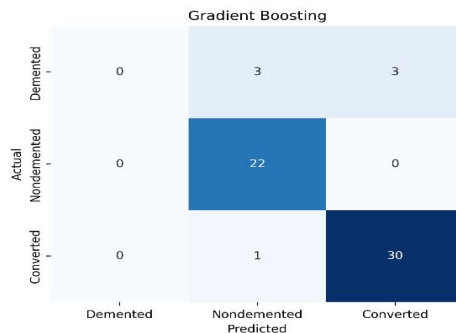
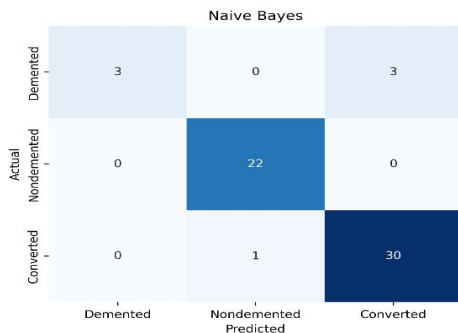
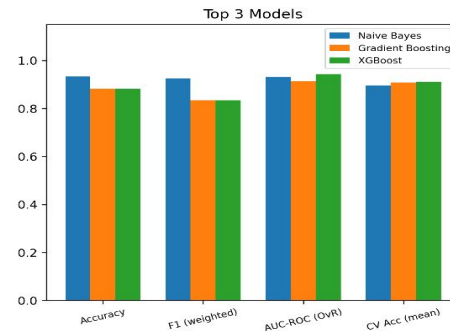
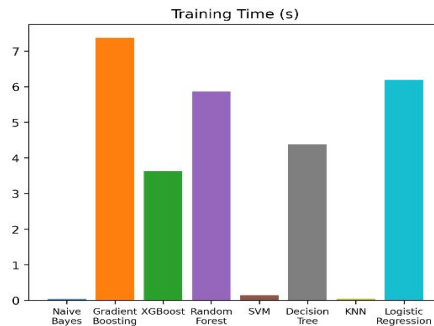
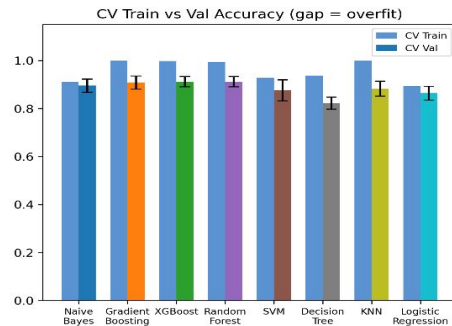
- Age & gender
- EDUC → Years of education completed.
- SES → Socioeconomic status score.
- MMSE → Mini-Mental State Examination score (0–30); higher = better cognitive function.
- CDR → Clinical Dementia Rating score; higher = more severe dementia.
- eTIV → Estimated Total Intracranial Volume (total skull/brain cavity size).
- nWBV → Normalized Whole Brain Volume; proportion of brain volume (lower may indicate brain atrophy).
- ASF → Atlas Scaling Factor; scaling value used to normalize brain size to a standard atlas.



Classical ML Results

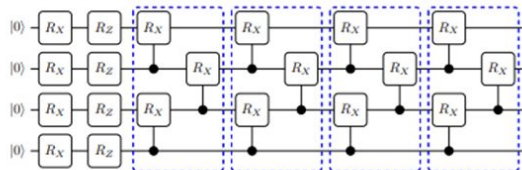
Model	Accuracy	F1 (weighted)	AUC-ROC	CV Acc (mean±std)	Train Time (s)
Naive Bayes	0.9322	0.9250	0.9313	0.8949 ± 0.0276	0.039
Gradient Boosting	0.8814	0.8344	0.9141	0.9078 ± 0.0271	7.38
XGBoost	0.8814	0.8344	0.9428	0.9109 ± 0.0213	3.626
Random Forest	0.8814	0.8344	0.9491	0.9109 ± 0.0213	5.872
SVM	0.8475	0.8555	0.9491	0.8756 ± 0.0440	0.148
Decision Tree	0.8475	0.8535	0.9141	0.8218 ± 0.0248	4.375
KNN	0.8305	0.7880	0.8614	0.8821 ± 0.0312	0.044
Logistic Regression	0.7966	0.8198	0.9713	0.8632 ± 0.0290	6.187

Classical ML Results

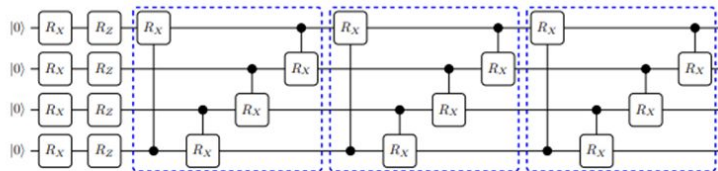


Quantum model selection Criteria : Entanglement, Expressiveness, & Time

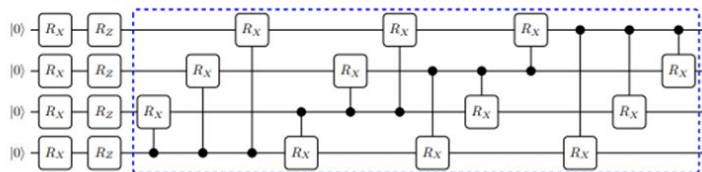
a) Nearest-neighbor (NN)



b) Circuit-block (CB)



c) All-to-all (AA)



1. Preferring CRx as it shows high expressiveness, and entangling capabilities
2. Applying different entanglement Pattern
3. Considering computational time, using circuit depth of 1 layer.

Entanglement Design	Relative Performance
NN (linear)	Baseline
Ring	Improved
All-to-all	Best performance

VQC1: CNOT linear chain

VQC2: CRX linear chain

VQC3: CRX ring

VQC4: CRX all-to-all

Entanglement Designs Tested:

Quantum VQC Architecture

① Angle Encoding

- 10 features $\rightarrow$ 10 qubits
- Map each feature $x_i \rightarrow RY(x_i)$ rotation
- $|\psi_i\rangle = RY(x_i)|0\rangle$
- Puts data directly into qubit angles
- Simulated Using PennyLane

VQC1: CNOT linear chain

② Variational Layer

- $RY(\theta) + RZ(\theta)$ per qubit - trainable
- Entangling gates connect qubits
- 4 designs vary entanglement topology
- Optimised via Adam (PennyLane)
- $Lr=0.01$,
- Mini-batch= 16

VQC2: CRX linear chain

③ Measurement

- $\langle \text{PauliZ} \rangle$ expectation on 3 qubits (3 classes)
- Softmax activation $\rightarrow$ 3-class probabilities
- Weighted cross entropy loss::
$$L = -\sum_k y_k \log(p_k)$$
- Class weighting applied to handle dataset imbalance

VQC3: CRX ring

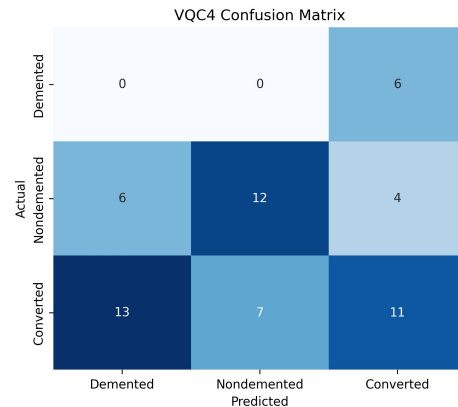
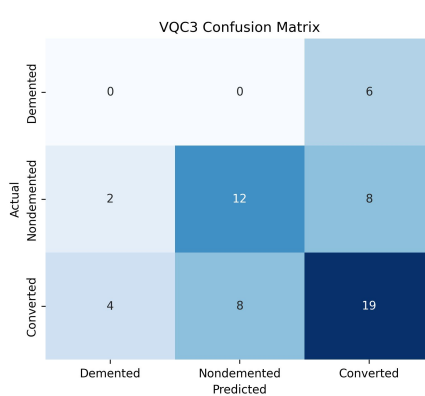
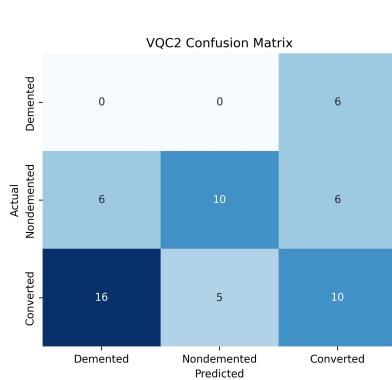
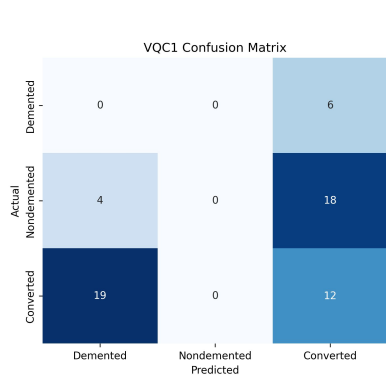
VQC4: CRX all-to-all

Gates per VQC Model (10 qubits) with single variational layer:

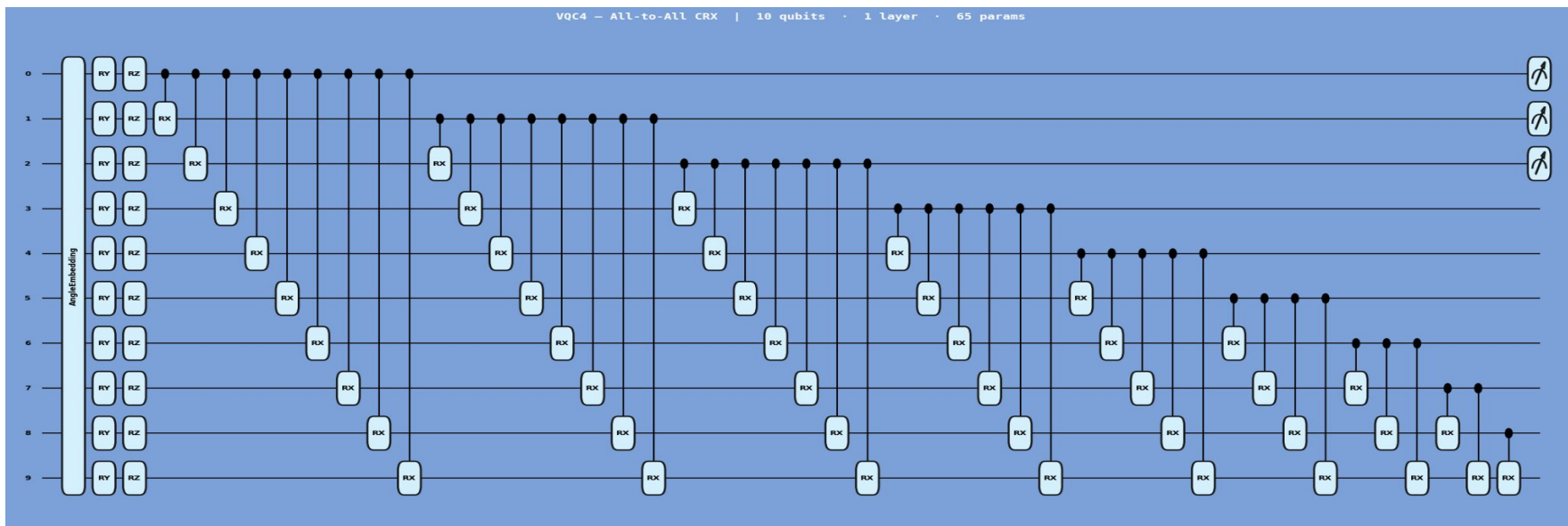
Model	Encoding	Variational Rotations	Entanglement	Connectivity	Trainable Params	Total Gates
VQC1	10 RY	10 RY + 10 RZ	9 CNOT	Linear	30	39
VQC2	10 RY	10 RY + 10 RZ	9 CRX	Linear	30	39
VQC3	10 RY	10 RY + 10 RZ	10 CRX	Ring	30	40
VQC4	10 RY	10 RY + 10 RZ	45 CRX	All-to-All	30	75

Quantum VQC Results

Model	Accuracy	F1	AUC-ROC	CV Acc (mean $\pm$ std)	Best loss	Train time (s)
VQC1 - CNOT linear	0.2034	0.1882	0.5138	0.4015 $\pm$ 0.0740	1.0212	670.7
VQC2 - CRX linear	0.3390	0.3998	0.4953	0.4711 $\pm$ 0.0625	0.9728	652.1
VQC3 - CRX ring	0.5254	0.5250	0.6953	0.4490 $\pm$ 0.0396	0.9914	839.1
VQC4 - CRX all-all	0.3898	0.4406	0.5891	0.4808 $\pm$ 0.0264	0.9512	6510.51



Tuned VQC4 (CRX All-to-All) with 3 layers, changeable Lr, more epochs



- The trainable block closes with 45 trainable CRX (Controlled-RX) gates applied to every unique qubit pair, creating full all-to-all entanglement across the register.
- In the full model, this layer is repeated 3 times with data re-uploading at each repetition, giving the circuit richer expressibility without increasing the qubit count.
- The final measurement takes the Pauli-Z expectation value on qubits 0, 1, and 2, producing three continuous outputs that are passed to a softmax for 3-class classification. Total trainable parameters: 195 (65 per layer × 3 layers).

Tuned VQC4 (CRX All-to-All) Pending Results:

— VQC4 | 5-Fold CV —

Fold 1/5

Epoch 001/40 | Loss: 1.3250 | LR: 0.01000

Epoch 002/40 | Loss: 1.2445 | LR: 0.01000

Epoch 003/40 | Loss: 1.1554 | LR: 0.01000

Epoch 039/40 | Loss: 0.6975 | LR: 0.00500

Epoch 040/40 | Loss: 0.6960 | LR: 0.00500

Fold 1 Val Acc: 0.7460

Epoch 039/40 | Loss: 0.7407 | LR: 0.00500

Epoch 040/40 | Loss: 0.7533 | LR: 0.00500

Fold 2 Val Acc: 0.7460

Epoch 035/40 | Loss: 0.7607 | LR: 0.00500

Epoch 036/40 | Loss: 0.7584 | LR: 0.00500

✓ Early stopping (CV)

Fold 3 Val Acc: 0.7778

Epoch 039/40 | Loss: 0.7167 | LR: 0.00500

Epoch 040/40 | Loss: 0.7200 | LR: 0.00500

Fold 4 Val Acc: 0.7778

- The 77.78% CV accuracy already outperform every completed VQC and is closing the gap with classical models like KNN (88.21% CV)
- The consistent accuracy across folds 1 & 2 (exactly 0.7460) suggests stable learning.

Classical vs Quantum — Why Quantum Matters

Best classical accuracy

93.2%

Naive Bayes

Best quantum accuracy

52.5%

VQC3 — CRX ring

Accuracy gap

−40.7pp

quantum vs classical

Fastest classical

0.04 s

vs ~652–6511 s quantum

Performance gap is large but expected

Classical models outperform all quantum models by 40+ percentage points. This is consistent with NISQ-era expectations.

Entanglement topology matters

VQC3 (ring) outperforms both the linear chain (VQC2) and the full all-to-all (VQC4) on test accuracy. This suggests that richer connectivity doesn't always help

Tuned VQC early results are highly promising

The tuned VQC4 variant (VQC5) shows 74.6% CV accuracy in early folds — a 22pp jump over VQC3 and narrowing the gap with classical models to ~19pp.

Training cost is a major constraint

VQC4 took ~6511 s (~1.8 hrs) vs under 8 s for any classical model. The all-to-all entanglement (45 CRX gates) makes simulation expensive on CPU.

Business Analysis & Future Work



Business Value in Healthcare



Screening Tool

Deploy in radiology clinics using existing MRI infrastructure — no new hardware needed



Cost Reduction

Early diagnosis cuts long-term care costs by up to 40% per patient



Clinical ROI

Converted group detection = highest value — catch patients before dementia onset



Future-proof

Quantum pipeline ready for quantum-enabled hospital infrastructure



Future Work

- Deeper VQC: add more variational layers for richer feature representations
- Softmax output layer for true class probabilities & better AUC-ROC
- Run on real quantum hardware (IBM Quantum / IonQ) to validate simulation results
- Extend to 3D MRI scans with quantum-enhanced CNN feature extraction

Key Takeaways



Naïve Bayes achieved 93.3% accuracy — best classical model for dementia early detection



VQC3 - CRX ring reached 52.9% — trainable entanglement is the key quantum design choice



Classical wins today, but quantum VQCs scale differently — advantage grows with data dimensionality



CDR and MMSE are the strongest predictors — clinically aligned, interpretable, and business-ready